

AI STUDYBUDDY API

PROJECT DOCUMENTATION

Project Title: "AI StudyBuddy API"

TEAM MEMBERS:

1. M.PRIYADHARSHINI • Team Leader
2. T.SRUTHI
3. C.SHARMILA
4. I.NASRIN BEGUM

INTRODUCTION

AI StudyBuddy API is a backend application developed to provide AI-powered learning support for students. It allows users to register, log in, upload study materials, and generate summaries, flashcards, quizzes, and study plans using the **Google Gemini API**. The application uses **MongoDB** for storing user data and study-related information.

PROJECT SETUP

The project was developed using **Visual Studio Code**. The required packages were installed using the following command:

npm install

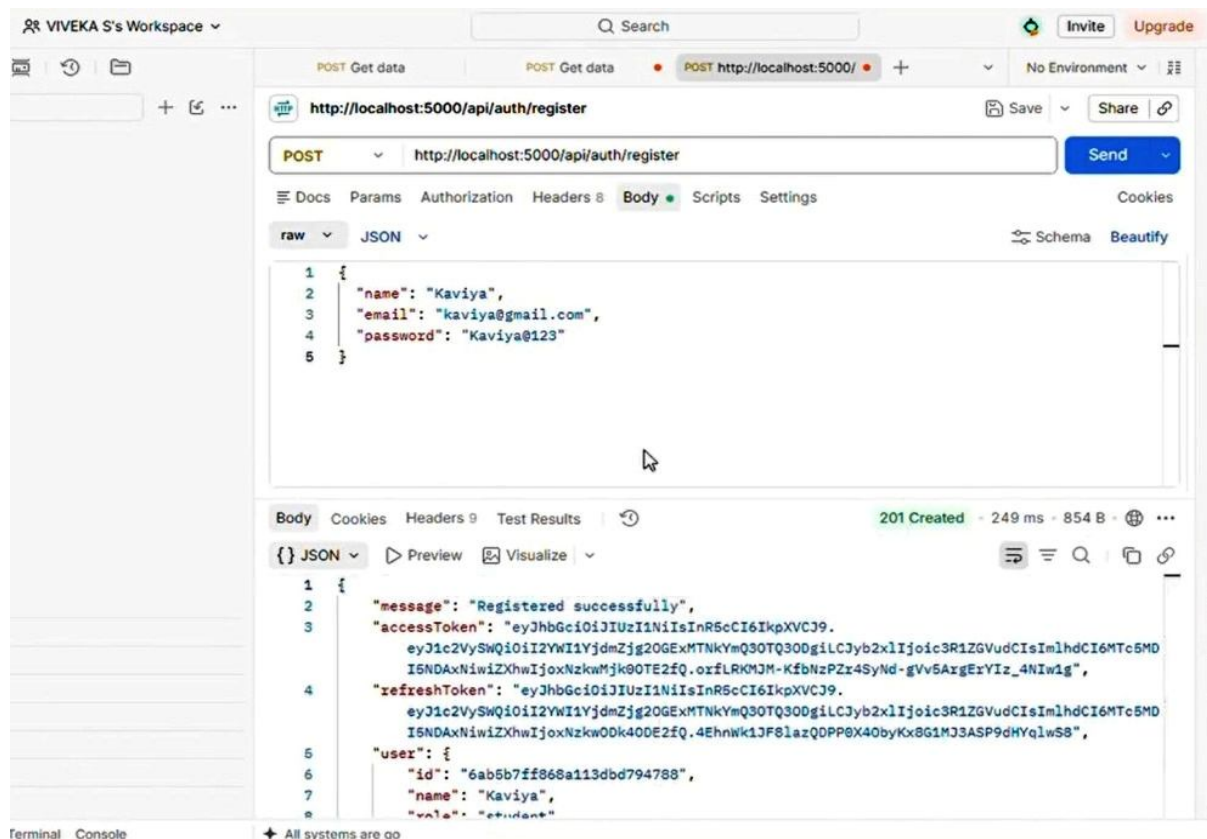
The backend application was started using:

npm start

API IMPLEMENTATION SCREENSHOTS

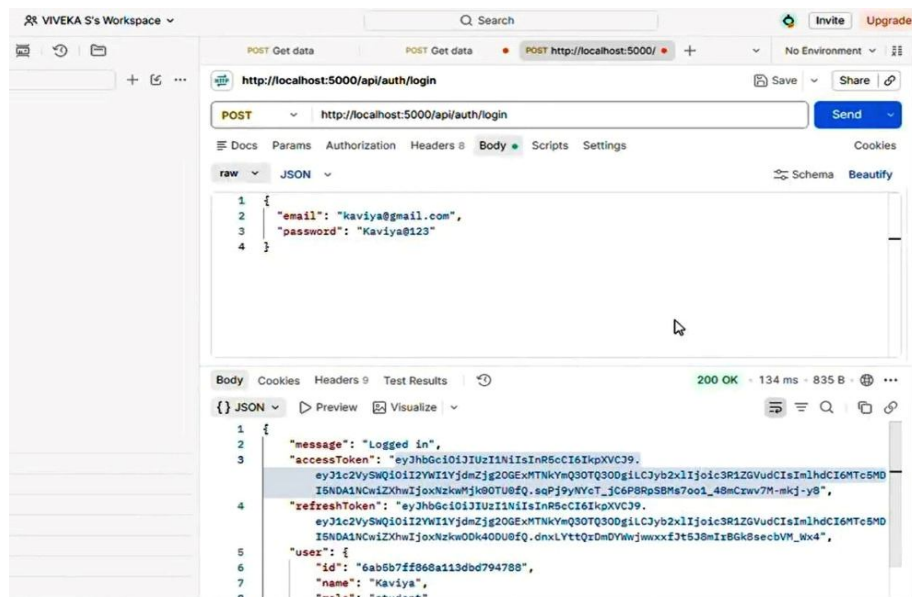
The following screenshots demonstrate the successful execution of **AI StudyBuddy API** using **Thunder Client in Visual Studio Code**, showing the API requests and corresponding server responses.

1. REGISTER API



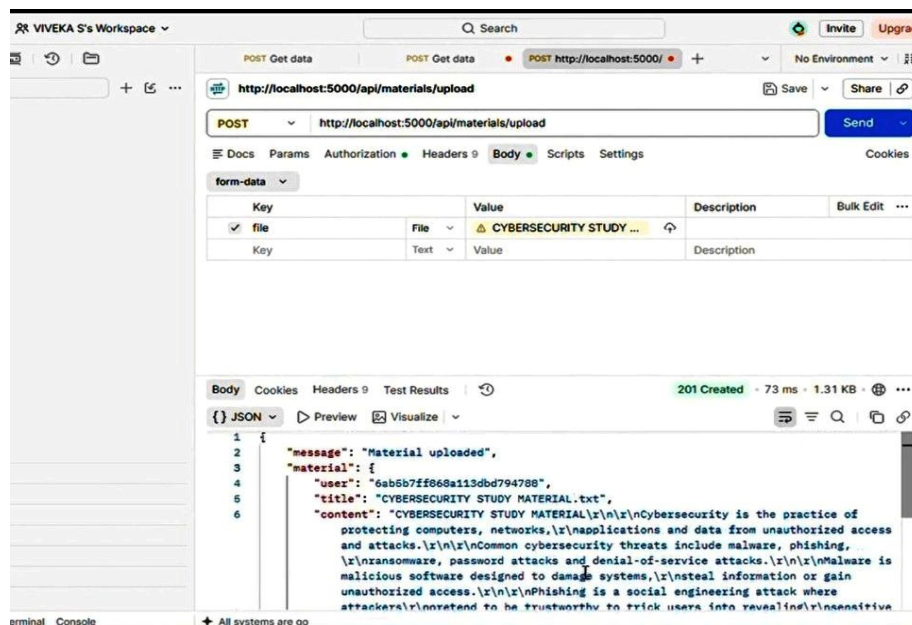
- This API allows a new user to **create an account** in the Study Buddy system.
- The user provides the required **registration details** in the request body.
- After successful registration, the user details are **stored in the MongoDB database**.

2. LOGIN API



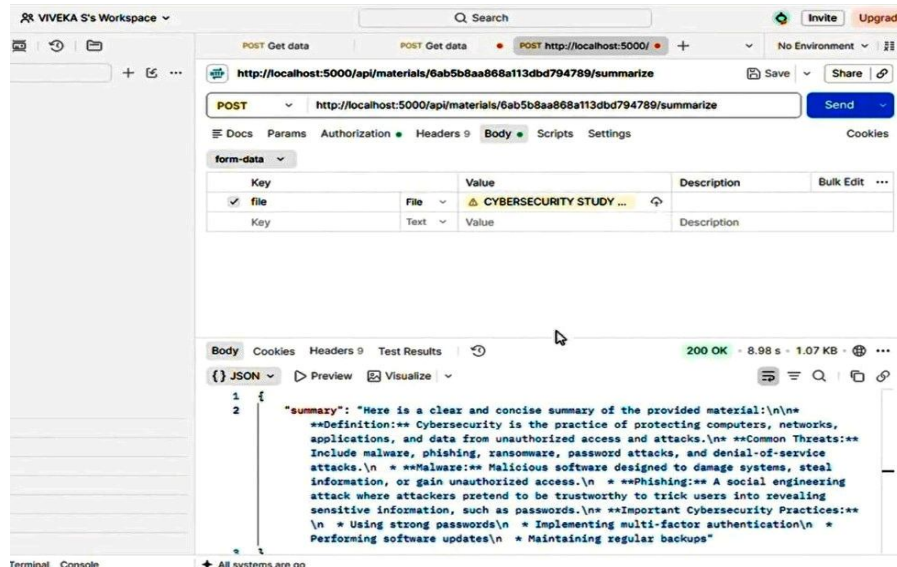
- This API allows an existing user to **log in to the Study Buddy system**.
- The user provides their **email and password** in the request body.
- If the credentials are valid, the system provides **authenticated access** to the application.

3. UPLOAD MATERIAL API



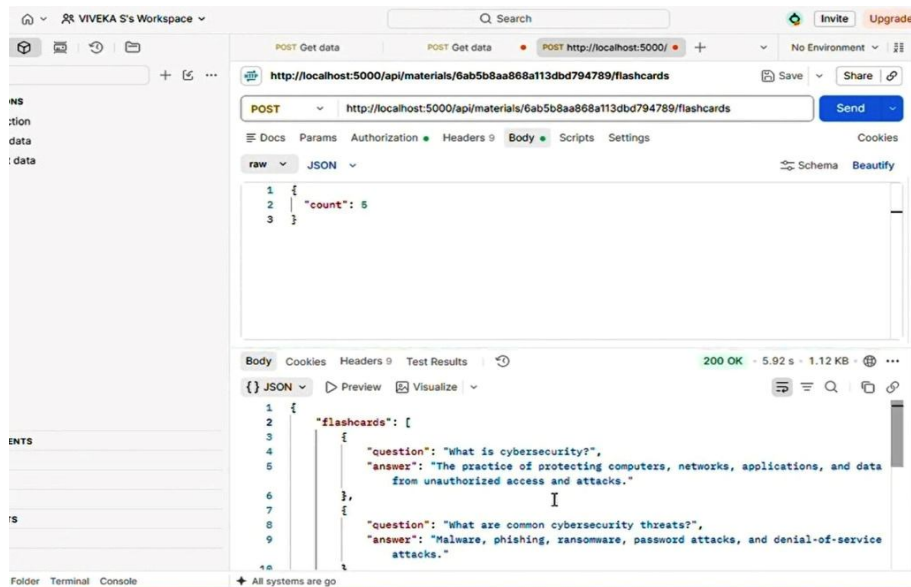
- This API allows the authenticated user to **upload study materials** to the system.
- The uploaded material is processed by the backend for further use.
- The material can then be used for **AI-powered learning features**.

4. SUMMARY API



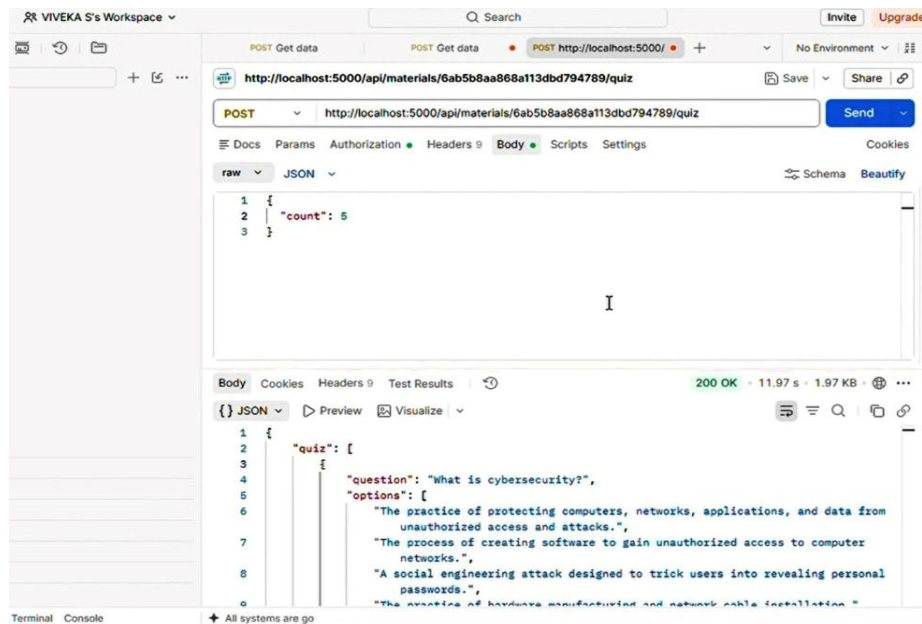
- This API generates a **summary of the uploaded study material**.
- The study material is processed using the **Google Gemini API**.
- The system returns a **concise AI-generated summary** to the user.

5. FLASHCARDS API



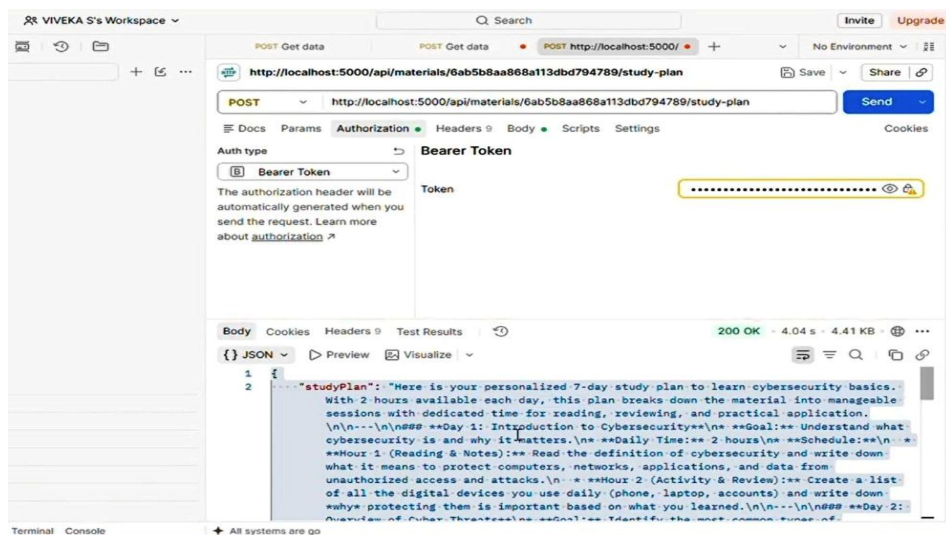
- This API generates **flashcards from the study material**.
- The content is processed using the **Google Gemini API**.
- The system returns **AI-generated flashcards** for study and revision.

6. QUIZ API



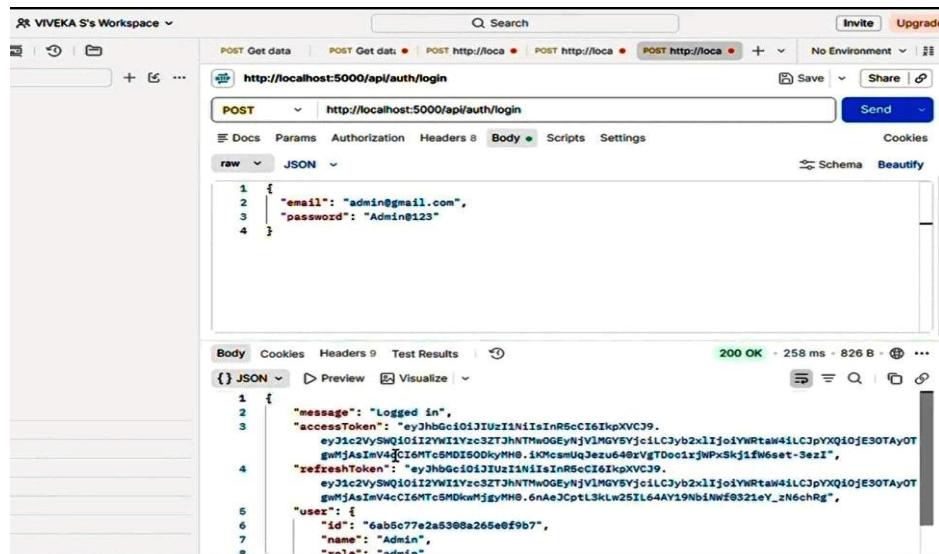
- This API generates a **quiz based on the uploaded study material**.
- The study content is processed using the **Google Gemini API**.
- The system returns **AI-generated quiz questions** for the user.

7. STUDY PLAN API



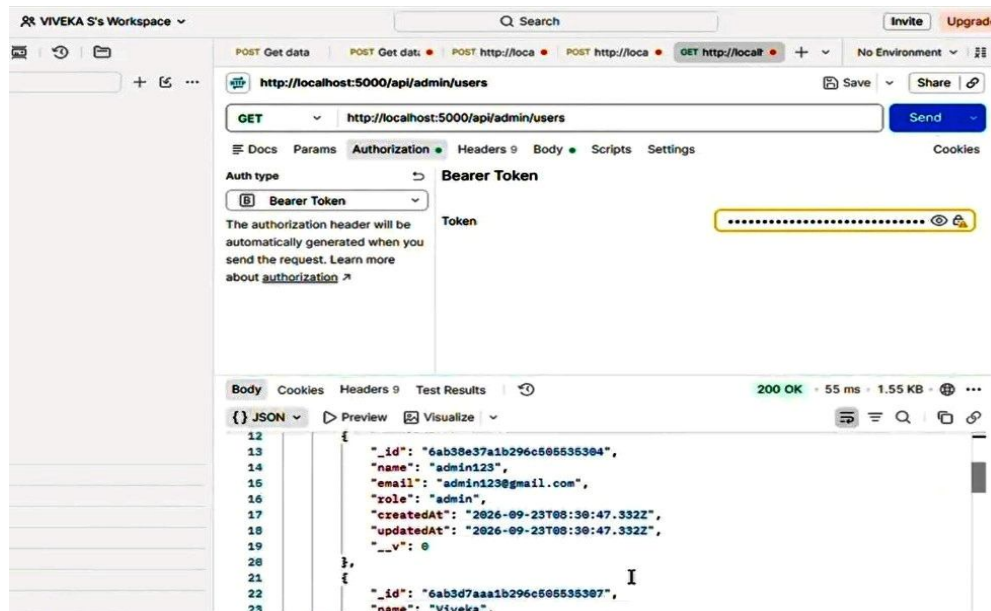
- This API helps the user **generate a personalized study plan**.
- The required study information is processed using the **Google Gemini API**.
- The system returns a **structured AI-generated study plan**.

8. ADMIN LOGIN API



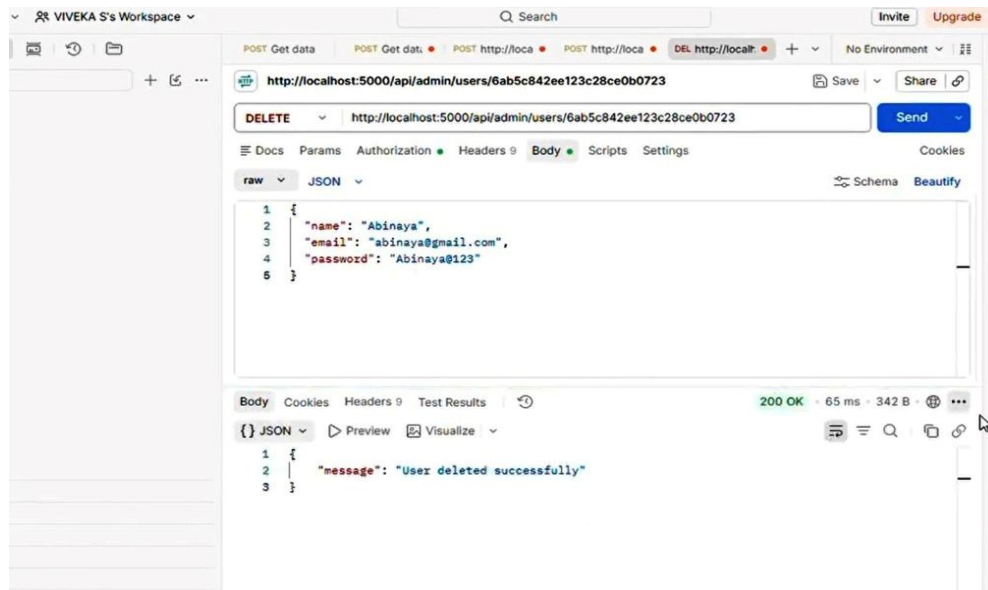
- This API allows the **administrator to log in** to the Study Buddy system.
- The administrator provides the required **login credentials**.
- After successful authentication, the administrator can **access user management features**.

9. GET ALL USERS API



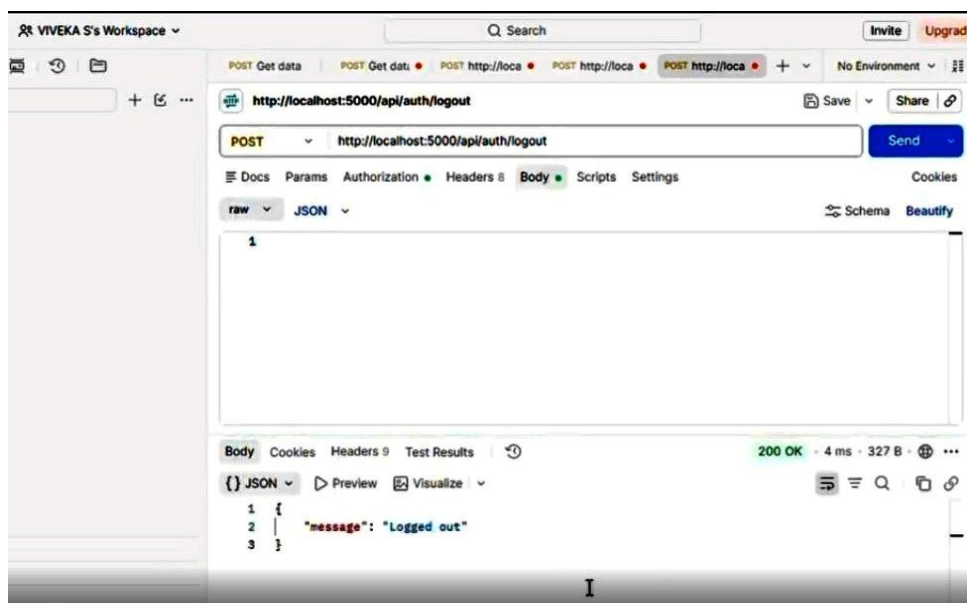
- This API allows the administrator to **retrieve the details of registered users**.
- The user information is fetched from the **MongoDB database**.
- The system returns the **available user details** in the response.

10. DELETE USER API



- This API allows the administrator to **delete a registered user** from the system.
- The user is identified using their **unique user ID**.
- After successful deletion, the system returns a **confirmation response**.

11. LOGOUT API



- This functionality allows the user to **log out from the Study Buddy system**.
- It ends the user's current **authenticated session**.
- The user must authenticate again to access protected features.